

Building a Parsimonious Model to Understand Human Life Expectancy in a Country

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Introduction

This project is on finding a parsimonious model to predict human life expectancy in a country. Country-level data was used to carry out the analysis that contains average life expectancy (in years) as the outcome, and the predictors include Land area (in km²), Population (in millions), percentage of population living in rural areas, percentage of government expenditures directed towards health care, percentage of population with internet access, number of births per 1000 people, percentage of population at least 65 years old, CO2 emissions in metric tons per capita, GDP per capita, cell phone subscriptions per 100 people.

Executive Summary

To construct the regression model, the relationship between each predictor and the outcome variable was analyzed using the scatter plot. Then, simple linear regressions with each predictor were performed to assess the regression assumptions between the predictor and outcome variable. To fix the issues of non-linearity and violations in normality, a complete multiple linear regression containing all possible predictors was done and transformations were done on the outcome variable and on the predictors. With the transformed outcome variable and predictors, a parsimonious model was found by an approximate model selection, regression diagnostics, assessing model assumptions, and multicollinearity. Finally, we found that land area, internet access, GDP, and birth rate to be significant predictors of life expectancy.

Model Building

To construct our model, we first analyzed the relationship between each predictor and our outcome variable, life expectancy. We did this by plotting a scatter plot of each predictor against life expectancy, noting any nonlinear behaviors or extraneous values. We found that land area, population of the country, internet access, each country's elderly population, each country's carbon emissions, GDP, and cell phones showed a nonlinear relationship to life expectancy. Simple linear regressions were performed to confirm that nonlinearity existed between these predictors and the outcome variable. It was also noted from the simple regressions that almost all of the normalized Q-Q plots showed violations of the normality assumption. Oftentimes, predictor variables can interact with each other in such a way that performing a multiple linear

regression can actually fix issues of nonlinearity and violations in normality. Thus, we first chose to do a complete multiple linear regression containing all possible predictors.

Our full multiple linear regression model indicated that a country's access to the internet, elderly population, birth rate, and number of cell phones were significant predictors of life expectancy at the 95% confidence level. We performed regression diagnostics on our full model to confirm that no regression assumptions had been violated. The regression diagnostics showed that nonlinear trends still existed and that the tails of the Q-Q plot clearly deviated from normality. Thus, we chose to transform both the outcome variable, life expectancy, and several of our predictor variables.

A Box-Cox transformation was performed to determine the appropriate transformation of our outcome variable. Table 1 shows an example of a list of the optimal lambda values for several predictors given from the Box-Cox transformation, and a list of the transformations performed to several of the independent variables. A lambda value of 4 was chosen for the transformation of the outcome variable.

	Population	Internet Access	Elderly Population	GDP	Cell Phones
Optimal λ Value	3.2	5.3	4	5.5	3.1
X-Transformation	$\log(x)$	$\sqrt{x}$	$\log(x)$	$\log(x)$	$\log(x)$

Table 1: A list of several predictors' optimal transformations and a range of optimal lambda values given by the Box-Cox transformation.

After the chosen transformation were found and performed, scatter plots of the newly transformed predictors against life expectancy were created to confirm that the nonlinearity issue had been resolved. The transformed predictors were then regressed against the newly transformed outcome variable to see if the transformations had altered which predictors were significant. We found that it had. Our full multiple linear regression model now reported that land area, population, internet access, birth rate, and GDP were significant at the 95% confidence level. To achieve a more parsimonious model, we used forward selection, backward selection, and best subsets model selections to help us narrow down our predictors to only include the most relevant. Finally, the potential for multicollinearity was assessed to help us understand the limitations and full applications of our model.

Results

From our model building process, we found two multiple linear regression models that contain solely predictors that are significant at the 95% confidence level. Both models show that the GDP, the birth rate of the country, and the percentage of the population with internet access are significant predictors of life expectancy, with one model also including the land area and the other including the population of the country. We chose to use the model that excludes population, as we are looking for a parsimonious model to understand life expectancy, and the model with land area had the highest values of both adjusted- R^2 ($\text{adj}R^2 = 0.818$) and Mallows's C_p ($C_p = 8.34$). Figure 1 shows the relationship between each significant predictor and life expectancy. Since these plots do not account for the interactions between predictors, a table of the regression coefficients has been included, as well.

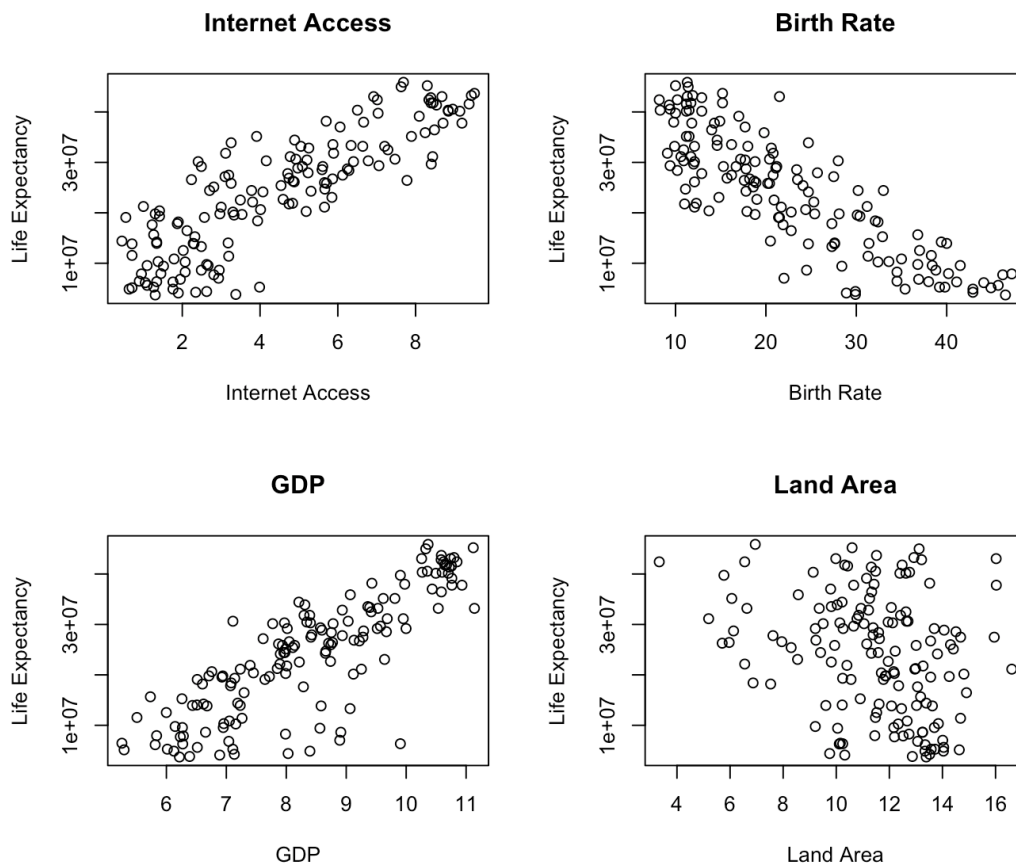


Figure 1: Scatter plots of relevant predictor variables versus the outcome life expectancy. Linear relationships clearly exist for each predictor and life expectancy.

Predictors of Life Expectancy	Beta Coefficients	95% Confidence Intervals
Internet Access	0.3883	(0.24977, 0.52690)
Birth Rate	-0.3635	(-0.48325, -0.24373)
GDP	0.2005	(0.05989, 0.34116)
Land Area	-0.07354	(-0.14532, -0.00176)

Table 2: Table showing the significant predictors of life expectancy included in the model. Values have been standardized for easier interpretation.

As we can see in Table 2, internet access and the GDP of the country are positively correlated with life expectancy. Birth rate of the countries and land area are negatively correlated. In this case, our predictors with the highest significance also correspond to the order of variables that contribute the most to predicting life expectancy. Internet access contributes the most to our model, followed by birth rate, GDP, and land area. Our model shows a strong relationship between internet access, birth rate, GDP, and land area with life expectancy, which can clearly be seen with the reported adjusted $R^2=0.818$. Figure 2 confirms that our linear model is valid by verifying that our regression assumptions are upheld.

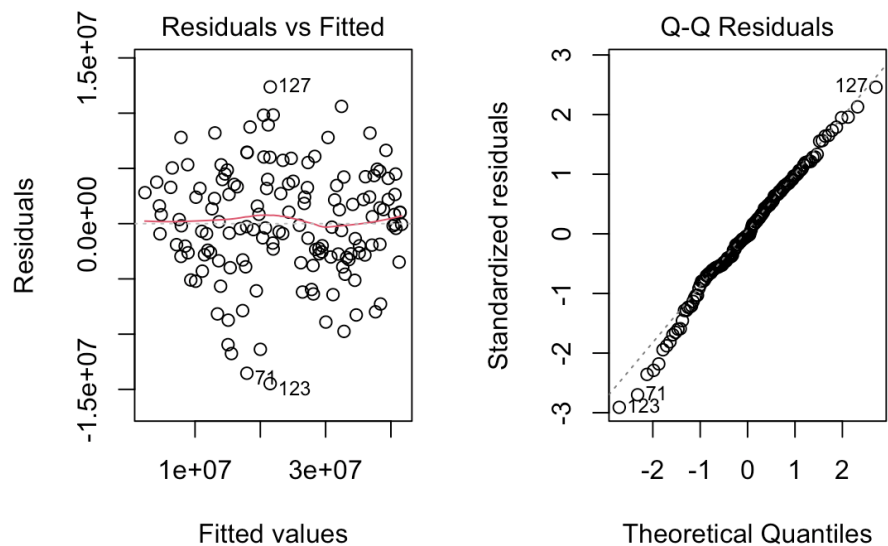


Figure 2: Regression diagnostics for our final model on life expectancy. The left plot shows the residuals versus the fitted values, while the right contains the normalized Q-Qplot. Regression assumptions are upheld.

In addition to that, multicollinearity was assessed using the correlation coefficients of all the variables and was found that GDP and internet had a very high correlation of 0.804 and also a significant correlation between internet and birth, birth and GDP, and internet and GDP (Refer to the appendix of the multicollinearity plot). Furthermore, VIF values (given in brackets) were determined for all the predictors and it was found that Internet (5.05), GDP (3.74), Birth rate (4.037) are moderately collinear but Area (1.2) is less collinear with other predictors. Figure 3 shows a visual representation of the relationship our predictors have to each other. Among the model predictors, before transformations, the Internet has the highest VIF value of 3.92, followed by GDP, birth rate and area. Area is the least correlated and Internet is moderately correlated with other model predictors. After transformations, VIF values have increased a bit, which shows that the correlation has increased. GDP has the highest VIF value of 4.09, which is not so different from that of the Internet (3.97), and followed by birth rate and area. Hence, GDP and Internet are moderately correlated whereas birth rate is less correlated, and area is the least correlated.

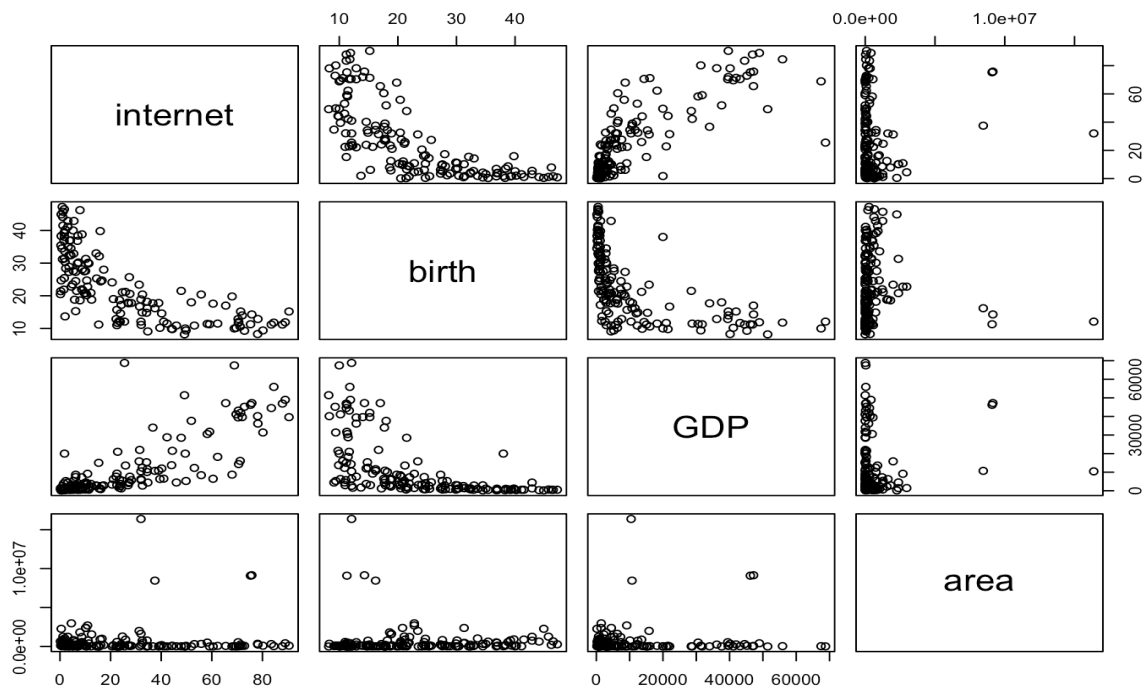


Figure 3: Multicollinearity plot showing significant predictors' relationships to each other.

Discussion and Conclusion

We found a parsimonious model to predict life expectancy that named internet access, GDP of each country, birth rate, and land area as significant predictors at the 95% confidence level of a country's life expectancy rate. Our final model showed no violations of our regression assumptions, but we did find a moderate collinear relationship between our predictors. Future work would entail investigating the effect multicollinearity has on our final model's applications to predicting life expectancy of a country.

Appendix

Appendix containing R code and additional plots included as a separate PDF. Multicollinearity scatter plot showing all the predictors' relationships to each other is given below.

